Results

77 out of 102 patients were treated with interstitial implants (no. of implants ranging 3-21, median of 6) and 25 by a mold technique. The 5-year control rates were 95%, 91% and 83% for local, regional and local-regional control respectively. Since the introduction of image guided brachytherapy in 2010, 5-year local control was 100% (n = 42). Tumor volume ≥2.3 cm^3 resulted in worse 3-year regional control (61%) compared to volume <2.3cm^3 (96%; p = .01). Ultimate regional control after salvage treatment was 96%, with no significant difference between subgroups by tumor volume (92% for ≥2.3 cm^3 vs 96% for <2.3 cm^3; p = .57). Patient assessed cosmetic and functional satisfaction were both rated high (mean 3.7 and 4.0 out of 5 respectively).

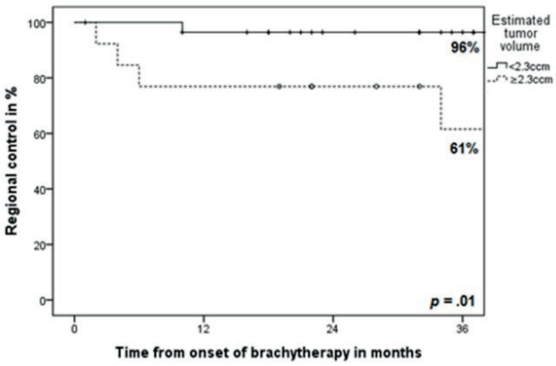

Figure 2: 3-year regional control by tumor volume (Log-rank test, p = .01).

Conclusion

We report the largest patient cohort treated with brachytherapy as the sole treatment for nasal vestibule carcinoma to date. Brachytherapy offers excellent local and regional control for Wang T1-T2 tumors with high patient satisfaction. Tumor volume is an adequate predictive factor for patients at risk of regional recurrence, but ultimate control rates after salvage treatment are high. Therefore, we do not recommend elective treatment of the neck. In future work, we want to explore sentinel node biopsy as diagnostic tool for high-risk patients with nasal vestibule carcinoma.

OC-0290 A predictive model for visus loss after uveal melanoma interventional radiotherapy (brachytherapy)

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Purpose or Objective

The aim of this study was to develop a predictive model on visus loss after ruthenium-106 plaque brachytherapy in uveal melanoma.

Material and Methods

Patients from institutional database with choroidal melanoma treated with ruthenium-106 plaque from December 2006 to November 2016 were selected. In each case the prescribed dose was 100 Gy at tumor apex. Factors analyzed were Age at diagnosis, Tumor thickness, Longitudinal tumor diameter, Transversal tumor diameter, Distance to optic disc, Gender, Quadrant, Localization, Morphology, Pigmentation, Dose fovea, Dose disco, Distance to the fovea, Diabetes, Hypertension. Univariate statistical tests (Pearson's Chi square and Two Sample t-test) and multivariate logistic regression were used to define the impact of baseline patient factors on the visus loss. A p-value <= 0.05 was considered significant. The model was chosen through stepwise selection on significative features at univariate analysis, and its performance was evaluated with internal cross-validation using Area Under the ROC Curve (AUC) and calibration with Hosmer-Lemeshow test.

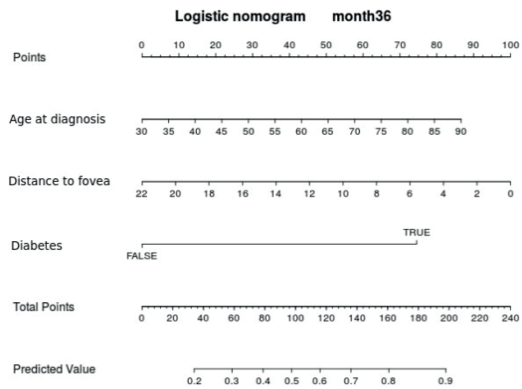
Results

187 patients with a median age of 65 years (range: 17-87) were considered for this analysis. The median follow-up was 72.25 months. Of 187 patients, 93% were alive. Visus loss was found in 98 patients (52.4%) at 3 year after the treatment. Distance to fovea was the main prognostic factor of the predictive model (odds ratio of 0.905 [0.845-0.965] p = 1.5e-03). Other prognostic factors were Age at diagnosis (odds ratio of 1.03 [1.01-1.05] p = 0.025) and Diabetes (odds ratio 5.10 [3.50-6.70] p=0.039) (table 1) . Based on this analysis we developed a nomogram (fig.1), with an AUC of 0.73, useful in the clinical practice. The calibration show no statistical difference between actual and predicted 3-year visus loss (p=0.28).

Feature	Odds ratio	p-value
Age at diagnosis	1.03	0.025
Distance to fovea	0.905	1.5e-03
Diabetes	5.10	0.039

Conclusion

Our prognostication model could be a tool for predicting visus loss at 3 years after treatment. Indeed, this analysis revealed that distance to the fovea, age and diabetes can help to predict visus loss at 3 years after treatment: a predictive model is provided and reasonable performance (AUC) encourage further investigations along this direction.



OC-0291 3D image-guided treatment planning of Ru-106 brachytherapy for choroidal melanomas

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